

Compositional reinforcement learning: a categorical approach

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1 Reinforcement learning introduction

1.1 What is Reinforcement Learning

Reinforcement learning is simultaneously a problem, a class of solution methods that work well on that class of problems, and the field that studies these problems and their solution methods. Reinforcement learning problems involve learning what to do so as to maximize a numerical reward signal; each such problem has three important features, which characterize this type of problem:

1. The problems are *closed-loop*, which means that the learning system's actions influence its later inputs.
2. The learner is not told which actions to take, as in many forms of machine learning, but instead must discover which actions yield the most reward by trying them out.
3. Actions may affect not only the immediate reward but also the next situation and, through that, all subsequent rewards.

The basic idea is simply to capture the most important aspects of the real problem facing a learning agent interacting with its environment to achieve a goal. Clearly, such an agent must be able to sense the state of the environment and must be able to take actions that affect the state. The agent also must have a goal or goals relating to the state of the environment. The formulation is intended to include just these three aspects—sensation, action, and goal—in their simplest possible forms without trivializing any of them.

1.2 Differences from Other Types of Learning

Supervised learning is learning from a training set of labeled examples provided by a knowledgeable external supervisor. Each example is a description of a situation together with a specification—the label—of the correct action the system should take in that situation, which is often to identify a category to which the situation belongs. The object of this kind of learning is for the system to extrapolate, or generalize, its responses so that it acts correctly in situations not present in the training set. This is an important kind of learning, but alone it is not adequate for learning from interaction. In reinforcement learning, an agent must be able to learn from its own experience.

Reinforcement learning is also different from what machine learning researchers call *unsupervised learning*, which is typically about finding structure hidden in collections of unlabeled data—data for which we do not know which examples are “good” and which are “bad”. Although one might be tempted to think of reinforcement learning as a kind of unsupervised learning because it does not rely on examples of correct behavior, reinforcement learning is trying to maximize a reward signal instead of trying to find hidden structure. Uncovering structure by itself does not address the reinforcement learning agent's problem of maximizing a reward signal.

1.3 Further features of reinforcement learning

One of the challenges that arise in reinforcement learning, and not in other kinds of learning, is the trade-off between exploration and exploitation. To obtain a large amount of reward, a reinforcement learning agent must prefer actions that it has tried in the past and found to be effective in producing reward. But to discover such actions, it has to try actions that it has not selected before. The agent has to exploit what it already knows in order to obtain reward, but it also has to explore in order to make better action selections in the future. The dilemma is that neither exploration nor exploitation can be pursued exclusively without failing at the task.

1.4 Examples

A good way to understand reinforcement learning is to consider some of the examples and possible applications that have guided its development:

- A master chess player makes a move. The choice is informed both by planning—anticipating possible replies and counterreplies—and by immediate, intuitive judgments of the desirability of particular positions and moves.
- A mobile robot decides whether it should enter a new room in search of more trash to collect or start trying to find its way back to its battery recharging station. It makes its decision based on the current charge level of its battery and how quickly and easily it has been able to find the recharger in the past.

All of the previous examples involve interaction between an active decision-making agent and its environment, within which the agent seeks to achieve a goal despite uncertainty about its environment. The agent's actions are permitted to affect the future state of the environment, thereby affecting the options and opportunities available to the agent at later times. At the same time, in all these examples the effects of actions cannot be fully predicted; thus the agent must monitor its environment frequently and react appropriately.

1.5 Elements of reinforcement learning

What we would like, ideally, is a state signal that summarizes past sensations compactly, yet in such a way that all relevant information is retained. This normally requires more than the immediate sensations, but never more than the complete history of all past sensations. A state signal that succeeds in retaining all relevant information is said to be Markov, or to have the *Markov property* (we define this formally below). For example, a checkers position—the current configuration of all the pieces on the board—would serve as a Markov state because it summarizes everything important about the complete sequence of positions that led to it.

In this section we want to define in a rigorous way all the ideas explained previously. We also have to say that one can choose a more or less general degree of generality to give these definitions. For example, since we want to analyze only environments with the Markov

property, our definition of policy will depend only on the current state. For less generality one could also consider the deterministic definition for a policy ¹ Moreover, we will consider time as a discrete variable, indexed by the non-negative integers.

For all the rest of the document, we will consider $\mathcal{A}$ (the set of actions) and $\mathcal{S}$ (the set of states) as two measurable spaces with σ -algebras $\mathcal{F}_{\mathcal{A}}$ and $\mathcal{F}_{\mathcal{S}}$ respectively. Moreover, we will consider only the case where $\mathcal{F}_{\mathcal{A}} = \mathcal{P}(\mathcal{A})$ is the power set of $\mathcal{A}$.

Definition 1.1 (Stochastic policy). A *stochastic policy* is a collection of maps

$$(\pi_t)_{t \in \mathbb{N}},$$

where, for every $t \in \mathbb{N}$,

$$\pi_t : \mathcal{S} \rightarrow \mathfrak{P}(\mathcal{A}),$$

and $\mathfrak{P}(\mathcal{A})$ denotes the set of probability measures on $(\mathcal{A}, \mathcal{P}(\mathcal{A}))$. For every state $s \in \mathcal{S}$, the probability of choosing an action in $\{a\} \in \mathcal{P}(\mathcal{A})$ at time t is denoted by

$$\pi_t(\{a\} \mid s).$$

Thus, a policy defines the learning agent's way of behaving at a given time. Now we want to define the most important random variables.

Definition 1.2. Let $(\mathcal{S}, \mathcal{F}_{\mathcal{S}})$ and $(\mathcal{A}, \mathcal{P}(\mathcal{A}))$ be measurable spaces, let $\mu_0 \in \mathfrak{P}(\mathcal{S}, \mathcal{F}_{\mathcal{S}})$ be an initial state distribution, and let P be a transition kernel. The *trajectory space* is the product space

$$\Omega := (\mathcal{S} \times \mathcal{A})^{\mathbb{N}},$$

equipped with the product σ -algebra

$$\mathcal{F}_{\Omega} := \bigotimes_{t \in \mathbb{N}} (\mathcal{F}_{\mathcal{S}} \otimes \mathcal{P}(\mathcal{A})).$$

For a fixed policy $(\pi_t)_{t \in \mathbb{N}}$, the probability measure $\mathbb{P}^{\mu_0, \pi, P}$ on $(\Omega, \mathcal{F}_{\Omega})$ is the unique measure given by Theorem B.18. The *canonical projection maps* are defined by

$$S_t(\omega) := s_t, \quad A_t(\omega) := a_t, \quad \omega = (s_0, a_0, s_1, a_1, \dots) \in \Omega.$$

Definition 1.3 (Reward signal). Let $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be a bounded $(\mathcal{F}_{\mathcal{S}} \otimes \mathcal{P}(\mathcal{A}))$ -measurable function, called the *reward function*. The *reward signal* is the sequence of random variables

$$R_{t+1} := r(S_t, A_t), \quad t \in \mathbb{N}.$$

The idea behind this definition is that at each time step $t \in \mathbb{N}$, the environment sends to the agent a single number $R_t \in \mathbb{R}$ (the reward). The agent's objective is to maximize the total reward accumulated over the long run. The reward signal thus determines what constitutes good and bad events for the agent.

We have said that the agent's goal is to maximize the cumulative reward it receives in the long run. How might this be defined formally?

¹A *deterministic policy* is a collection of maps $(\pi_t)_{t \in \mathbb{N}} : \mathcal{S} \rightarrow \mathcal{A}$.

Definition 1.4 (Value Function). Fix $\gamma \in [0, 1)$, for any s the map $\delta_s \in \mathfrak{P}(\mathcal{S}, \mathcal{F}_S)$, a transition kernel P , and a policy $(\pi_t)_{t \in \mathbb{N}}$. The *value function* under $(\pi_t)_{t \in \mathbb{N}}$ is the map

$$v^\pi : \mathcal{S} \rightarrow \mathbb{R}, \quad v^\pi(s) := \mathbb{E}^{\delta_s, \pi, P} \left[\sum_{k=0}^{\infty} \gamma^k R_{k+1} \right],$$

where γ is called the *discount factor*.

Whereas the reward signal indicates what is good in an immediate sense, a *value function* specifies what is good in the long run. While rewards determine the immediate, intrinsic desirability of states, values capture the long-term desirability of states, taking into account the likely successor states and the rewards available in those states.

Remark 1.5. The intuition behind the discount factor γ is the following: the closer γ is to 0, the greater the emphasis placed on immediate rewards; conversely, the closer γ is to 1, the greater the importance assigned to long-term rewards.

Remark 1.6. The value function is well defined: since r is bounded by some $M > 0$ and $\gamma \in [0, 1)$, the series $\sum_{k=0}^{\infty} \gamma^k R_{k+1}$ converges absolutely $\mathbb{P}^{\delta_s, \pi, P}$ -almost surely, with

$$\sum_{k=0}^{\infty} \gamma^k |R_{k+1}(\omega)| \leq \frac{M}{1 - \gamma} < \infty.$$

The exchange of sum and expectation is justified by the dominated convergence theorem.

Value functions define a quasi partial ordering over policies². A policy π is defined to be better than or equal to a policy π' if its expected return is greater than or equal to that of π' for all states. In other words, $\pi \geq \pi'$ if and only if $v_\pi(s) \geq v_{\pi'}(s)$ for all $s \in \mathcal{S}$. There is always at least one policy that is better than or equal to all other policies, this is an *optimal policy*. Although there may be more than one, we denote all the optimal policies by π_* ³. They share the same state-value function, called the *optimal state-value function*, denoted v_* , and defined as

$$v_*(s) = \max_{\pi} v_\pi(s) \tag{1}$$

for all $s \in \mathcal{S}$.

²Indeed, the order is a relation that is not antisymmetric.

³Since the order is not anti-symmetric, we have that if also holds $v_\pi = v_{\pi'}$ we can have that $\pi \neq \pi'$.

2 Markov decision process

2.1 The category MDP

Definition 2.1 (MDPs). A MDP is a 5-tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \psi, T, R)$ consisting of:

- the *state space* $\mathcal{S}$, a numerable measurable space with a σ -algebra $\mathcal{F}_{\mathcal{S}}$;
- the *action space* $\mathcal{A}$, a numerable measurable space with the σ -algebra $\mathcal{P}(\mathcal{A})$;
- a function $\psi : \mathcal{A} \rightarrow \mathcal{S}$ that maps an action $a \in \mathcal{A}$ to its associated state $s \in \mathcal{S}$. Every action in the action space $a \in \mathcal{A}$ can be taken at a specific state $s \in \mathcal{S}$, and ψ maps the action to that state so that $\psi(a) = s$. Equivalently, the set of actions available at s is the preimage $\psi^{-1}(s) \subseteq \mathcal{A}$;
- the transition probabilities $T : \mathcal{A} \rightarrow \mathfrak{P}_{\mathcal{S}}$, where $\mathfrak{P}_{\mathcal{S}}$ denotes the space of probability measures on $\mathcal{S}$;
- the function $R : \mathcal{A} \rightarrow \mathbb{R}$.

Remark 2.2. Using the map T , we can define a transition kernel map (definition B.17) as follows:

$$\begin{aligned} P : \mathcal{S} \times \mathcal{A} \times \mathcal{F}_{\mathcal{S}} &: \rightarrow [0, 1] \\ (s, a, B) &\mapsto T(a)(B) \end{aligned}$$

Thus, fixing a MDP implies fixing also a transition kernel map.

Remark 2.3. Let $\mathcal{M}$ be a MDP. We equip $\mathcal{A}$ with the discrete σ -algebra $\mathcal{P}(\mathcal{A})$. Using the function R we can define the reward function

$$\begin{aligned} r : \mathcal{S} \times \mathcal{A} &\rightarrow \mathbb{R} \\ (s, a) &\mapsto R(a). \end{aligned}$$

Note that, since every action $a \in \mathcal{A}$ is associated to a unique state $\psi(a) \in \mathcal{S}$, the reward $R(a)$ is effectively defined on pairs $(\psi(a), a) \in \mathcal{S} \times \mathcal{A}$. Thus, we also have a reward signal $(R_t)_{t \in \mathbb{N}}$ linked to the MDP $\mathcal{M}$, together with the notions of return and optimal policy introduced above.

Definition 2.4. An *optimal policy* for an MDP $\mathcal{M}$ is the optimal policy with respect to the reward signal defined above.

Remark 2.5. Let us spell out the interpretation of the definitions above.

- The state space $\mathcal{S}$ and the action space $\mathcal{A}$ represent the possible states and actions available to the agent.
- The function ψ encodes the availability of actions: an action $a \in \mathcal{A}$ is available at state $s \in \mathcal{S}$ if and only if $\psi(a) = s$, so that the set of actions available at s is precisely the preimage $\psi^{-1}(s)$.

- The function R assigns a numerical reward to each action, and is used to define the reward signal as described in the remark above.
- The transition probabilities T encode the "stochastic dynamics of the environment": given an action $a \in \mathcal{A}$ available at state $s = \psi(a)$, the probability measure $T(a) \in \mathfrak{P}_{\mathcal{S}}$ describes the distribution over next states $s' \in \mathcal{S}$ that the agent will transition to after taking action a .

Definition 2.6 (Category of MDPs). MDPs form a category, called **MDP**, whose objects are MDPs and whose morphisms are as follows. Let $\mathcal{M}_i = (\mathcal{S}_i, \mathcal{A}_i, \psi_i, T_i, R_i)$, with $i = 1, 2$, be two MDPs. A morphism $m = (f, g) : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ between two MDPs is the data of a measurable function $f : \mathcal{S}_1 \rightarrow \mathcal{S}_2$ and a function $g : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ satisfying the following compatibility conditions:

- The diagram

$$\begin{array}{ccc} \mathcal{A}_1 & \xrightarrow{g} & \mathcal{A}_2 \\ \downarrow \psi_1 & & \downarrow \psi_2 \\ \mathcal{S}_1 & \xrightarrow{f} & \mathcal{S}_2 \end{array} \quad (2)$$

commutes.

- The diagram

$$\begin{array}{ccc} \mathcal{A}_1 & \xrightarrow{g} & \mathcal{A}_2 \\ \downarrow T_1 & & \downarrow T_2 \\ \mathfrak{P}_{\mathcal{S}_1} & \xrightarrow{f_*} & \mathfrak{P}_{\mathcal{S}_2} \end{array} \quad (3)$$

commutes.

- The diagram

$$\begin{array}{ccc} \mathcal{A}_1 & \xrightarrow{g} & \mathcal{A}_2 \\ & \searrow R_1 & \downarrow R_2 \\ & & \mathbb{R} \end{array} \quad (4)$$

commutes, i.e. $R_1 = R_2 \circ g$.

Definition 2.7 (Subprocess). We say that the MDP $\mathcal{M}_1$ is a *subprocess* of the MDP $\mathcal{M}_2$ if there exists a morphism $m = (f, g) : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ where f and g are injective, that is, the morphism m is injective on state and action spaces. Moreover, we say that a subprocess $\mathcal{M}_1$ is a *full subprocess* of $\mathcal{M}_2$ if diagram (2) is cartesian in the category **Set**, i.e.

$$\mathcal{A}_1 \cong \mathcal{S}_1 \prod_{\mathcal{S}_2} \mathcal{A}_2 = \{(s, a) \in \mathcal{S}_1 \times \mathcal{A}_2 \mid f(s) = \psi_2(a)\}$$

Remark 2.8. Let $\mathcal{N}$ be a full subprocess of $\mathcal{M}$, with morphism (f, g) where f and g are inclusions. From the full subprocess condition, we get that

$$\mathcal{A}_{\mathcal{N}} \cong \mathcal{S}_{\mathcal{N}} \prod_{\mathcal{S}_{\mathcal{M}}} \mathcal{A}_{\mathcal{M}} = \{(s, a) \in \mathcal{S}_{\mathcal{M}} \times \mathcal{A}_{\mathcal{M}} \mid f(s) = \psi(a)\} = \{(s, a) \in \mathcal{S}_{\mathcal{M}} \times \mathcal{A}_{\mathcal{M}} \mid s = \psi(a)\}$$

where the last equality holds since f is the inclusion. Moreover, we have the following commutative diagram

$$\begin{array}{ccc} \mathcal{A}_{\mathcal{N}} & \xrightarrow{g} & \mathcal{A}_{\mathcal{M}} \\ \downarrow \psi_{\mathcal{N}} & & \downarrow \psi_{\mathcal{M}} \\ \mathcal{S}_{\mathcal{N}} & \xrightarrow{f} & \mathcal{S}_{\mathcal{M}} \end{array}$$

Fix $s \in \mathcal{S}_{\mathcal{N}}$. If $a \in (\mathcal{A}_{\mathcal{N}})_s$, then $a \in \mathcal{A}_{\mathcal{M}}$ and $\psi_{\mathcal{M}}(a) = f(s)$, so a is an action available at $f(s)$ in $\mathcal{M}$ as well. Conversely, for any $a \in (\mathcal{A}_{\mathcal{M}})_{f(s)}$, we have $(s, a) \in \mathcal{S}_{\mathcal{N}} \times_{\mathcal{S}_{\mathcal{M}}} \mathcal{A}_{\mathcal{M}}$. By the universal property of the fiber product, there exists a unique $\bar{a} \in \mathcal{A}_{\mathcal{N}}$ such that $g(\bar{a}) = a$. Since g is injective, $\bar{a} = a$, hence $a \in \mathcal{A}_{\mathcal{N}}$. Therefore, the actions available at s in $\mathcal{N}$ are in bijection with the actions available at $f(s)$ in $\mathcal{M}$. In other words, if an agent is in the environment $\mathcal{N}$, it can move using exactly the same actions available in $\mathcal{M}$ at the corresponding state, without leaving $\mathcal{N}$.

2.2 Coproducts in MDP

In this section we analyze pushouts in the category **MDP**. The category **MDP** also admits fiber products (see [GBT25]).

Theorem 2.9. Suppose that we have three MDPs $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ together with two morphisms $m_1 = (f_1, g_1): \mathcal{M}_3 \rightarrow \mathcal{M}_1$ and $m_2 = (f_2, g_2): \mathcal{M}_3 \rightarrow \mathcal{M}_2$. There exists an MDP $\mathcal{M} = \mathcal{M}_1 \coprod_{\mathcal{M}_3} \mathcal{M}_2$ and morphisms $\mathcal{M}_1 \rightarrow \mathcal{M}$, $\mathcal{M}_2 \rightarrow \mathcal{M}$ fitting in a pushout diagram in **MDP**:

$$\begin{array}{ccc} \mathcal{M}_3 & \xrightarrow{m_1} & \mathcal{M}_1 \\ m_2 \downarrow & & \downarrow \\ \mathcal{M}_2 & \longrightarrow & \mathcal{M} \end{array}$$

Proposition 2.10. Suppose that $\mathcal{M}_3$ is a subprocess of $\mathcal{M}_1$ and $\mathcal{M}_2$. Then $\mathcal{M}_1$ and $\mathcal{M}_2$ are subprocesses of $\mathcal{M}_1 \coprod_{\mathcal{M}_3} \mathcal{M}_2$.

2.3 The Bellman equation for MDPs

In general, fixing a policy π and a discount factor $\gamma \in [0, 1)$, one can obtain a more explicit formula for the value function. We state the theorem (without proof) in the particular case of MDPs as defined above.

Theorem 2.11 (Bellman optimality equation for MDPs). For any state $s \in \mathcal{S}$ in the state space of an MDP $\mathcal{M}$, the optimal value function satisfies

$$v_*(s) = \max_{a \in \mathcal{A}_s} \sum_{s' \in \mathcal{S}} T(a)(s') [R(a) + \gamma v_*(s')].$$

2.4 Zigzag diagrams

Definition 2.12 (Zigzag diagram). Let us consider a collection $\{\mathcal{M}_0, \dots, \mathcal{M}_n, \mathcal{N}_0, \dots, \mathcal{N}_{n-1}\}$ of MDPs such that $\mathcal{N}_i$ is a subprocess of both $\mathcal{M}_i$ and $\mathcal{M}_{i+1}$ for any $i \in \{0, \dots, n-1\}$. A *zigzag diagram* is a diagram in the category **MDP** of the form

$$\begin{array}{ccccccc} & \mathcal{N}_0 & & \mathcal{N}_1 & & \dots & & \mathcal{N}_{n-1} \\ & \swarrow \quad \searrow & & \swarrow \quad \searrow & & & & \swarrow \quad \searrow \\ \mathcal{M}_0 & & \mathcal{M}_1 & & \mathcal{M}_2 & \dots & \mathcal{M}_{n-1} & & \mathcal{M}_n \end{array} \quad (5)$$

Using the previous results, we can give the following definition.

Definition 2.13. Given a zigzag diagram D as in (5), for any $n \in \mathbb{N}$ we define the *composite* MDP $\mathcal{C}_n$ by the following inductive rule:

$$\begin{aligned} \mathcal{C}_0 &:= \mathcal{M}_0, \\ \mathcal{C}_1 &:= \mathcal{C}_0 \coprod_{\mathcal{N}_0} \mathcal{M}_1 = \mathcal{M}_0 \coprod_{\mathcal{N}_0} \mathcal{M}_1, \\ \mathcal{C}_2 &:= \mathcal{C}_1 \coprod_{\mathcal{N}_1} \mathcal{M}_2 = \left(\mathcal{M}_0 \coprod_{\mathcal{N}_0} \mathcal{M}_1 \right) \coprod_{\mathcal{N}_1} \mathcal{M}_2, \\ &\vdots \\ \mathcal{C}_n &:= \mathcal{C}_{n-1} \coprod_{\mathcal{N}_{n-1}} \mathcal{M}_n \end{aligned}$$

The above definition was designed to model compositional tasks.

Remark 2.14. What is the idea behind these definitions? Each subprocess $\mathcal{N}_i \rightarrow \mathcal{M}_i$ models the completion of a task in the sense that the goal of an agent is to eventually reach a state $s \in \mathcal{S}_{\mathcal{N}_i}$. Once the i -th goal is accomplished inside the environment $\mathcal{M}_i$, we allow for the possibility of a changing environment and more options for states and actions in order to achieve the next goal, modeled by the subprocess $\mathcal{N}_{i+1} \rightarrow \mathcal{M}_{i+1}$. The composite MDP $\mathcal{C}_n$ is a single environment capturing all the tasks simultaneously.

A formal way to express the previous remark is the following proposition.

Proposition 2.15. Given a zigzag diagram D as in (5), we have $\mathcal{C}_n^D = \text{colim}(D)$.

Now we want to analyze the following problem.

Question 2.16. Suppose an agent has learned an optimal policy for each MDP $\mathcal{M}_i$ given the reward function R_i for achieving the i -th goal, for each $i = 0, \dots, n$. Under what conditions do these optimal policies determine an optimal policy for the joint reward on the composite MDP $\mathcal{C}_n$?

In order to answer this question, we introduce some additional definitions.

Definition 2.17. Fix a discount factor $\gamma \in [0, 1)$ and a zigzag diagram as in (5). The optimal value function $v_*^{\mathcal{C}_n}$ is said to be *monotonic* if, for any $i \in \{0, \dots, n-1\}$ and for any state $s \in \mathcal{S}_i$, the quantities

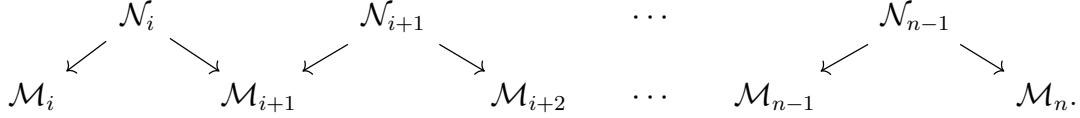
$$\sum_{s' \in \mathcal{S}_i} T(a)(s') (R_i(a) + \gamma \cdot v_*^{\mathcal{C}_n}(s'))$$

and

$$\sum_{s' \in \mathcal{S}_i} T(a)(s') (R_i(a) + \gamma \cdot v_*^{\mathcal{C}_{[i,n]}}(s'))$$

are maximized by the same action $a \in (\mathcal{A}_i)_s$.

In the previous definition, $\mathcal{C}_{[i,n]}$ denotes the composite MDP of the truncated zigzag diagram



Remark 2.18. Let us unpack what the definition is requiring. The optimal action in state $s \in \mathcal{S}_i$ is the one maximizing

$$\sum_{s' \in \mathcal{S}_i} T(a)(s') (R_i(a) + \gamma \cdot v_*(s')).$$

The definition asks that this optimal action is the same whether we use $v_*^{\mathcal{C}_n}$, the optimal value function of the full composite MDP $\mathcal{C}_n$, or $v_*^{\mathcal{C}_{[i,n]}}$, the optimal value function of the truncated composite MDP from $\mathcal{M}_i$ to $\mathcal{M}_n$. By the Bellman optimality equation (Theorem 2.11), this means that selecting a at s and continuing optimally thereafter realizes the optimal value in both chains. Since $\mathcal{C}_{[i,n]}$ is obtained from $\mathcal{C}_n$ by removing the tasks $\mathcal{M}_0, \dots, \mathcal{M}_{i-1}$ that precede $\mathcal{M}_i$, while the tasks $\mathcal{M}_{i+1}, \dots, \mathcal{M}_n$ that follow are left unchanged, we have that the optimal value obtainable from $\mathcal{M}_i$ onward, and the action realizing it, do not depend on which tasks come before $\mathcal{M}_i$.

Definition 2.19. A zigzag diagram D as in (5) is said to be *forward-moving* if $\mathcal{N}_i$ is a full subprocess of $\mathcal{M}_i$ for any $i \in \{0, \dots, n-1\}$.

One may wonder why the term “forward-moving” is appropriate. The following lemma provides the justification.

Lemma 2.20. If D as in (5) is a forward-moving zigzag diagram, then $\mathcal{C}_{[i+1,n]}$ is a full subprocess of $\mathcal{C}_{[i,n]}$.

Remark 2.21. By the previous lemma, $\mathcal{C}_{[i+1,n]}$ is a full subprocess of $\mathcal{C}_{[i,n]}$. By Remark 2.8, once an agent reaches a state of $\mathcal{C}_{[i+1,n]}$, it cannot leave it even using the actions of the larger environment $\mathcal{C}_{[i,n]}$. In other words, once the i -th task has been accomplished, the agent can only move forward to solve the subsequent tasks. This justifies the name of the property.

Theorem 2.22. Suppose that a zigzag diagram is forward-moving and the optimal value function of $\mathcal{C}_n$ is monotonic. Then, following the optimal policy π_i on each component $\mathcal{M}_i$ yields an optimal policy on the composite MDP $\mathcal{C}_n$.

Remark 2.23 (Every zigzag diagram can be made forward-moving). Although forward-motion is a genuine restriction on a given zigzag diagram, it is always possible to *replace* each $\mathcal{M}_i$ by a smaller MDP $\widehat{\mathcal{M}}_i$, obtained by removing exactly those actions that prevent $\mathcal{N}_i$ from being full, so that the resulting diagram becomes forward-moving. Concretely, given a sub-process $\mathcal{N}_i \hookrightarrow \mathcal{M}_i$ with state space $\mathcal{S}_{\mathcal{N}_i} \subseteq \mathcal{S}_i$ and action set $\mathcal{A}_{\mathcal{N}_i} \subseteq \psi_i^{-1}(\mathcal{S}_{\mathcal{N}_i})$, define

$$\widehat{\mathcal{M}}_i := (\mathcal{S}_i, \widehat{\mathcal{A}}_i, \psi_i, T_i, R_i) \Big|_{\widehat{\mathcal{A}}_i}, \quad \widehat{\mathcal{A}}_i := \mathcal{A}_i \setminus (\psi_i^{-1}(\mathcal{S}_{\mathcal{N}_i}) \setminus \mathcal{A}_{\mathcal{N}_i}),$$

i.e. we discard exactly those actions of $\mathcal{M}_i$ that originate in $\mathcal{S}_{\mathcal{N}_i}$ but are not already actions of $\mathcal{N}_i$. By construction,

$$\psi_i^{-1}(\mathcal{S}_{\mathcal{N}_i}) \cap \widehat{\mathcal{A}}_i = \mathcal{A}_{\mathcal{N}_i} = \mathcal{S}_{\mathcal{N}_i} \times_{\mathcal{S}_i} \widehat{\mathcal{A}}_i,$$

so the availability diagram of ψ_i , computed with respect to $\widehat{\mathcal{M}}_i$, is now cartesian: $\mathcal{N}_i$ is a full sub-process of $\widehat{\mathcal{M}}_i$.

Replacing every $\mathcal{M}_i$ by $\widehat{\mathcal{M}}_i$ in this way yields a new zigzag diagram $\widehat{D}$ that is forward-moving by construction.

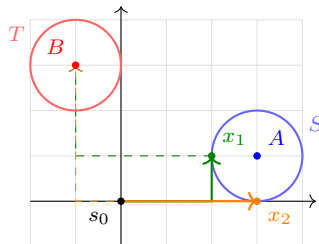
Two remarks are in order:

- this procedure only removes actions from $\mathcal{M}_i$, never changes the state space $\mathcal{S}_i$ nor the goal region $\mathcal{S}_{\mathcal{N}_i}$;
- removing actions can change the resulting composite MDP $\mathcal{C}_n$: an agent inside $\mathcal{S}_{\mathcal{N}_i}$ loses access to actions of $\mathcal{M}_i$ that were not already available in $\mathcal{N}_i$. Forward-motion is therefore obtained at the cost of restricting the agent's behavior at the gluing states, not for free.

Example 2.24 (Failure of monotonicity). Consider an agent moving on the grid $\mathbb{Z}^2$, starting from the origin $s_0 = (0, 0)$. At every state the agent may take one of the four unit-step actions $Step := \{\uparrow, \downarrow, \leftarrow, \rightarrow\}$, each moving it deterministically to an adjacent grid point.

Fix two target balls of radius 1 (with respect to the Euclidean norm) centered at $A = (3, 1)$ and $B = (-1, 3)$, denoted

$$\text{Ball}(A) := \{s \in \mathbb{Z}^2 : \|s - A\|_2 \leq 1\}, \quad \text{Ball}(B) := \{s \in \mathbb{Z}^2 : \|s - B\|_2 \leq 1\}.$$



We define three MDPs sharing the same state space $\mathcal{S} = \mathbb{Z}^2$ and the same deterministic transition dynamics described above, but differing in their reward functions.

$\mathcal{M}_0$:

$$\mathcal{M}_0 = (\mathcal{S}, \mathcal{A}, \psi, T, R_0), \quad \mathcal{S} = \mathbb{Z}^2$$

and

$$\mathcal{A} = (\mathcal{S} \times \{\uparrow, \downarrow, \leftarrow, \rightarrow\}) \setminus \{(x, \cdot) \mid \forall x \in \text{Ball}(A), \forall \cdot \in \text{Step}\}$$

⁴ where ψ sends an action (s, d) to its state of origin s , and T is

$$T : \mathcal{A} \rightarrow \mathfrak{P}_{\mathcal{S}}, \quad T(s, d) = \delta_{s+d}.$$

where $s + d$ is the point s moved by one in the verse of the arrow d . The reward function is

$$R_0(s, d) = \begin{cases} +1 & \text{if the resulting state lies in } \text{Ball}(A), \\ -1 & \text{otherwise.} \end{cases}$$

Thus $\mathcal{M}_0$ rewards the agent purely for reaching the neighborhood of A , with no information about what happens afterwards.

$\mathcal{N}_0$: $\mathcal{N}_0$ is the full sub-process of $\mathcal{M}_0$ whose state space is $\text{Ball}(A)$ itself, i.e.

$$\mathcal{S}_{\mathcal{N}_0} = \text{Ball}(A), \quad \mathcal{A}_{\mathcal{N}_0} = \psi^{-1}(\text{Ball}(A)),$$

with ψ, T, R_0 restricted accordingly. This represents the “goal region” of the first task, shared with the second task as the gluing locus of the zigzag diagram.

$\mathcal{M}_1$:

$$\mathcal{M}_1 = (\mathcal{S}, \mathcal{A}, \psi, T, R_1),$$

with the same $\mathcal{S}, \mathcal{A}, \psi, T$ as above, but reward

$$R_1(s, d) = \begin{cases} +1 & \text{if the resulting state lies in } \text{Ball}(B), \\ -1 & \text{otherwise.} \end{cases}$$

Thus $\mathcal{M}_1$ rewards the agent purely for reaching the neighborhood of B .

$\mathcal{C}_1$: Following the general construction, set

$$\mathcal{C}_0 := \mathcal{M}_0, \quad \mathcal{C}_1 := \mathcal{C}_0 \coprod_{\mathcal{N}_0} \mathcal{M}_1,$$

so that $\mathcal{C}_1$ models the full two-task episode: first reach $\text{Ball}(A)$, then, once inside it, proceed to reach $\text{Ball}(B)$.

⁴With this definition, we have that $\mathcal{M}_0$ is a full subprocess for $\mathcal{N}_0$ (just check the possible actions available in a state in $\text{Ball}(A)$ in $\mathcal{M}_0$).

Now consider two candidate paths from s_0 to $\text{Ball}(A)$, both of length 3 and both optimal for $\mathcal{M}_0$ in isolation (both reach the ball after the same number of steps and with the same reward):

$$\begin{aligned} s_0 &\rightarrow (1, 0) \rightarrow (2, 0) \rightarrow x_1 = (2, 1) \in \text{Ball}(A), \\ s_0 &\rightarrow (1, 0) \rightarrow (2, 0) \rightarrow x_2 = (3, 0) \in \text{Ball}(A). \end{aligned}$$

Since R_0 only depends on membership in $\text{Ball}(A)$, the optimal policy for $\mathcal{M}_0$ alone is indifferent between the two: both x_1 and x_2 are optimal terminal points of the first task.

However, once the second task is taken into account, the two choices are not equivalent: the shortest path from x_1 to $\text{Ball}(B)$ has length 5, while the shortest path from x_2 to $\text{Ball}(B)$ has length 7 (compare the solid and dashed paths in the picture). Hence

$$v_*^{\mathcal{C}_1}(x_1) > v_*^{\mathcal{C}_1}(x_2),$$

so the action leading to x_1 is strictly better than the one leading to x_2 once the composite task $\mathcal{C}_1$ is considered, even though the two actions are equally optimal for $\mathcal{M}_0$ in isolation.

3 A Compositional approach to Robot Manipulation

In this section we analyze a concrete experiment illustrating the theory developed in the previous sections, following [BSZ⁺25]. We consider a robot manipulation task in which a robot must move an object from one location to another. Rather than training a single RL agent on the full task, we apply the compositional framework described above: by Theorem 2.22, it suffices to find an optimal policy for each sub-task individually, and the resulting composite policy is guaranteed to be optimal for the full task.

We begin by giving formal definitions of the sub-task MDPs and their corresponding subprocesses.

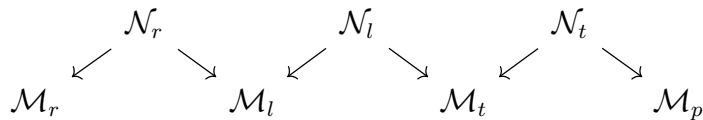
The four sub-task MDPs are defined as follows:

- $\mathcal{M}_r[r, o]$ models the *reaching task*, where the robot r must reach its grippers around an object o of interest. The 6-dimensional state space consists of the position of the robot hand and the object: $\mathbf{s} = [\mathbf{p}_r, \mathbf{p}_o]$. At each step, the robot controls the displacement of its hand, moving at most 10 cm in each direction: $\mathbf{a} = \Delta\mathbf{p}_r$. When the MDP transitions from state $\mathbf{s}$ to $\mathbf{s}'$, the robot receives a shaped reward $r = \|\mathbf{p}_r - \mathbf{p}_o\|_2 - \|\mathbf{p}'_r - \mathbf{p}'_o\|_2$.
- $\mathcal{M}_l[r, o]$ models the *lifting task*, assuming the robot hand is aligned with the object. The robot is only allowed up-and-down movements plus gripper controls. The state space contains the robot hand height, object height, and gripper width: $\mathbf{s} = [z_r, z_o, w]$, and $\mathbf{a} = [\Delta z_r, \Delta w]$. The robot is rewarded for staying close to the object and lifting it off the table: $r = (|z_r - z_o| - |z'_r - z'_o|) + (z'_r - z_r)$.
- $\mathcal{M}_t[r, o]$ is the *transporting task*. The robot must transport a held object to the other side of the table. The robot controls the hand movement but cannot open the gripper. Because the robot is holding the object, they share the same position, so $\mathbf{s} = \mathbf{p}_r$ and $\mathbf{a} = \Delta\mathbf{p}_r$. The robot is rewarded for moving in the $+Y$ direction: $r = y'_r - y_r$.
- $\mathcal{M}_p[r, o, g]$ is the *placing task*. Starting with the robot holding the object, it must place the object at the desired goal location g . The state space contains the robot position, object position, goal position, and gripper width: $\mathbf{s} = [\mathbf{p}_r, \mathbf{p}_o, \mathbf{p}_g, w]$. The robot controls the hand movement and gripper opening: $\mathbf{a} = [\Delta\mathbf{p}_r, \Delta w]$.

Next, we define the subprocesses corresponding to the sub-task MDPs. Each subprocess $\mathcal{N}_i$ is defined as the full sub-MDP of $\mathcal{M}_i$ obtained by restricting to the subset of states satisfying the transition condition below, inheriting the transition kernel and reward function from $\mathcal{M}_i$.

- $\mathcal{N}_r[r, o]$ contains the states where the robot position is at most 1 cm away from the object: $\|\mathbf{p}_r - \mathbf{p}_o\|_2 < 0.01$.
- $\mathcal{N}_l[r, o]$ is entered when the object's height is above some threshold h : $z_o > h$.
- $\mathcal{N}_t[r, o]$ contains the states where the robot has crossed the centre line of the table: $y_r > 0.03$.

We now consider the following zigzag diagram:



This diagram encodes the full manipulation task: the robot must reach the object, lift it, transport it across the table, and place it at the goal location. The MDPs $\mathcal{N}_i$ represent the transition conditions between sub-tasks, while the MDPs $\mathcal{M}_i$ represent the environments in which each sub-task is carried out.

In this experimental setting, following [BSZ⁺25], one can verify that the diagram is forward-moving — that is, each $\mathcal{N}_i$ is a full subprocess of $\mathcal{M}_i$ — and that the optimal value function of the composite MDP satisfies the monotonicity condition. Theorem 2.22 then guarantees that an optimal policy for the full task can be obtained by composing optimal policies for each individual sub-task. Consequently, one can train RL agents directly on the sub-tasks rather than on the full task.

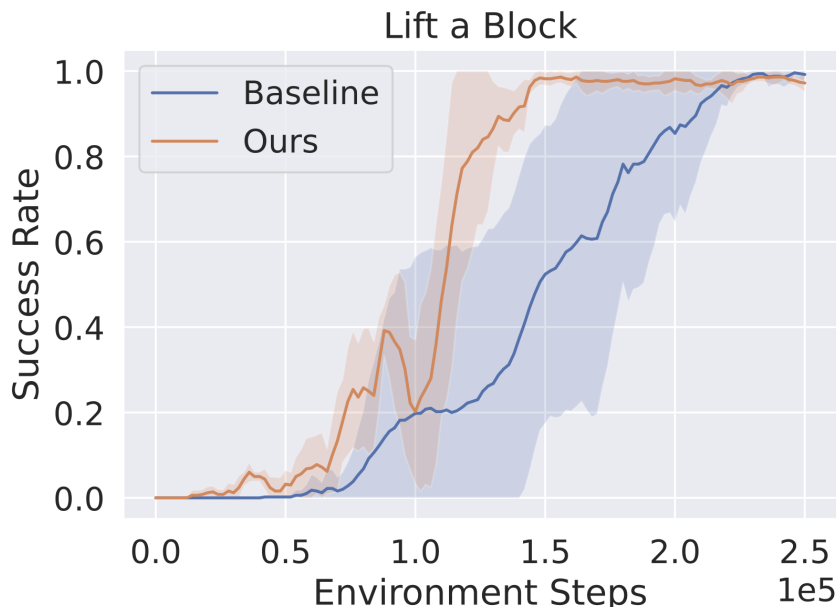


Figure 1: Success rate as a function of training steps for the compositional method and the single-agent baseline, reproduced from [BSZ⁺25]. The compositional approach reaches a 100% success rate after approximately 150k training steps, compared to 225k steps required by the baseline.

As shown in Figure 1, the experiments of [BSZ⁺25] demonstrate significantly better sample efficiency for the compositional method: it reaches a 100% success rate after approximately 150k training steps, whereas the baseline requires around 225k steps to achieve comparable performance.

The compositional approach also offers an additional practical advantage. Since complex tasks are decomposed into elementary sub-tasks, policies learned for one task can be reused

or fine-tuned for future tasks, even when the overall objective differs substantially. This avoids redundant training and reduces the total computational cost when deploying robots across a variety of related manipulation tasks.

A Category theory

A.1 Categories and functors

Definition A.1 (Category). A *category* $\mathbf{C}$ consists of:

- a collection $\text{ob}(\mathbf{C})$ of *objects*;
- for every pair of objects $X, Y \in \text{ob}(\mathbf{C})$, a set $\text{Hom}_{\mathbf{C}}(X, Y)$ of *morphisms* from X to Y ;
- for every triple of objects $X, Y, Z \in \text{ob}(\mathbf{C})$, a *composition map*

$$\circ : \text{Hom}_{\mathbf{C}}(Y, Z) \times \text{Hom}_{\mathbf{C}}(X, Y) \rightarrow \text{Hom}_{\mathbf{C}}(X, Z), \quad (g, f) \mapsto g \circ f;$$

satisfying the following axioms:

- *Associativity*: for all morphisms $f \in \text{Hom}(X, Y)$, $g \in \text{Hom}(Y, Z)$, $h \in \text{Hom}(Z, W)$,

$$h \circ (g \circ f) = (h \circ g) \circ f;$$

- *Identity*: for every object $X \in \text{ob}(\mathbf{C})$, there exists a morphism $\text{id}_X \in \text{Hom}_{\mathbf{C}}(X, X)$ such that for all $f \in \text{Hom}_{\mathbf{C}}(X, Y)$,

$$f \circ \text{id}_X = f = \text{id}_Y \circ f.$$

Example A.2. The following are standard examples of categories:

- **Set**: sets and functions;
- **Top**: topological spaces and continuous maps;
- **Ab**: abelian groups and group homomorphisms;
- **Meas**: measurable spaces and measurable maps.

Definition A.3 (Functor). Let $\mathbf{C}$ and $\mathbf{D}$ be categories. A *functor* $F : \mathbf{C} \rightarrow \mathbf{D}$ consists of:

- a map $F : \text{ob}(\mathbf{C}) \rightarrow \text{ob}(\mathbf{D})$;
- for every pair of objects $X, Y \in \text{ob}(\mathbf{C})$, a map

$$F : \text{Hom}_{\mathbf{C}}(X, Y) \rightarrow \text{Hom}_{\mathbf{D}}(F(X), F(Y));$$

satisfying the following axioms:

- *Preservation of identities*: for every $X \in \text{ob}(\mathbf{C})$,

$$F(\text{id}_X) = \text{id}_{F(X)};$$

- *Preservation of composition*: for all morphisms $f \in \text{Hom}_{\mathbf{C}}(X, Y)$ and $g \in \text{Hom}_{\mathbf{C}}(Y, Z)$,

$$F(g \circ f) = F(g) \circ F(f).$$

Example A.4. The following are standard examples of functors:

- the *forgetful functor* $\mathbf{Ab} \rightarrow \mathbf{Set}$, which sends a group to its underlying set and a homomorphism to the underlying function;
- the *fundamental group functor* $\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Grp}$, which sends a pointed topological space to its fundamental group.

A.2 Colimits

Definition A.5 (Diagram and colimit). Let $\mathbf{C}$ be a category and $\mathbf{J}$ a small category. A *diagram* of shape $\mathbf{J}$ in $\mathbf{C}$ is a functor $D : \mathbf{J} \rightarrow \mathbf{C}$. A *cocone* under D is an object $C \in \text{ob}(\mathbf{C})$ together with morphisms $\iota_j : D(j) \rightarrow C$ for every $j \in \text{ob}(\mathbf{J})$, such that for every morphism $\alpha : j \rightarrow j'$ in $\mathbf{J}$ the diagram

$$\begin{array}{ccc} D(j) & \xrightarrow{D(\alpha)} & D(j') \\ & \searrow \iota_j & \downarrow \iota_{j'} \\ & & C \end{array}$$

commutes. A *colimit* of D , denoted $\text{colim}(D)$, is a cocone $(C, (\iota_j)_j)$ that is universal, in the sense that for any other cocone $(C', (\iota'_j)_j)$ there exists a unique morphism $u : C \rightarrow C'$ such that $u \circ \iota_j = \iota'_j$ for all $j \in \text{ob}(\mathbf{J})$:

$$\begin{array}{ccc} D(j) & \xrightarrow{\iota_j} & C \\ & \searrow \iota'_j & \downarrow \exists! u \\ & & C' \end{array}$$

A.3 Pushouts and pullbacks

Definition A.6 (Pushout). Let $\mathbf{C}$ be a category and let $m_1 : Z \rightarrow X$ and $m_2 : Z \rightarrow Y$ be two morphisms with common domain. The *pushout* of m_1 and m_2 is an object $X \amalg_Z Y \in \text{ob}(\mathbf{C})$ together with morphisms $i_1 : X \rightarrow X \amalg_Z Y$ and $i_2 : Y \rightarrow X \amalg_Z Y$ such that $i_1 \circ m_1 = i_2 \circ m_2$, and which is universal with this property: for any object W and morphisms $f : X \rightarrow W$, $g : Y \rightarrow W$ with $f \circ m_1 = g \circ m_2$, there exists a unique morphism $u : X \amalg_Z Y \rightarrow W$ such that $u \circ i_1 = f$ and $u \circ i_2 = g$:

$$\begin{array}{ccccc} Z & \xrightarrow{m_1} & X & & \\ m_2 \downarrow & & \downarrow i_1 & \searrow f & \\ Y & \xrightarrow{i_2} & X \amalg_Z Y & \xrightarrow{\exists! u} & W \\ & \searrow g & & & \end{array}$$

The pushout is the colimit of the diagram $X \leftarrow Z \rightarrow Y$.

Example A.7. In the category **Set**, the pushout of two functions $m_1 : Z \rightarrow X$ and $m_2 : Z \rightarrow Y$ is the *amalgamated disjoint union*

$$X \coprod_Z Y := (X \sqcup Y) / \sim,$$

where $\sqcup$ denotes the disjoint union and $\sim$ is the equivalence relation generated by $m_1(z) \sim m_2(z)$ for all $z \in Z$. The canonical morphisms are the compositions

$$i_1 : X \hookrightarrow X \sqcup Y \twoheadrightarrow X \coprod_Z Y, \quad i_2 : Y \hookrightarrow X \sqcup Y \twoheadrightarrow X \coprod_Z Y.$$

In particular, when $Z = \emptyset$, the pushout reduces to the disjoint union $X \sqcup Y$.

Definition A.8 (Pullback). Let $\mathbf{C}$ be a category and let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be two morphisms with common codomain. The *pullback* of f and g is an object $X \times_Z Y \in \text{ob}(\mathbf{C})$ together with morphisms $p_1 : X \times_Z Y \rightarrow X$ and $p_2 : X \times_Z Y \rightarrow Y$ such that $f \circ p_1 = g \circ p_2$, and which is universal with this property: for any object W and morphisms $h : W \rightarrow X$, $k : W \rightarrow Y$ with $f \circ h = g \circ k$, there exists a unique morphism $u : W \rightarrow X \times_Z Y$ such that $p_1 \circ u = h$ and $p_2 \circ u = k$:

A commutative square is said to be *cartesian* if it is a pullback square. The pullback is the limit of the diagram $X \rightarrow Z \leftarrow Y$.

Example A.9. In the category **Set**, the pullback of two functions $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ is the *fiber product*

$$X \times_Z Y := \{(x, y) \in X \times Y \mid f(x) = g(y)\},$$

equipped with the projection maps $p_1(x, y) := x$ and $p_2(x, y) := y$.

B Measure theory and probability

B.1 Measurable spaces and σ -algebras

Definition B.1 (σ -algebra). Let Ω be a non-empty set. A family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is called a *σ -algebra* on Ω if:

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$ then $A^c := \Omega \setminus A \in \mathcal{F}$ (closure under complementation);

3. if $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$ then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ (closure under countable unions).

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*. Elements of $\mathcal{F}$ are called *measurable sets*, or *events* in a probabilistic context.

Remark B.2. From the three axioms one immediately deduces: $\emptyset \in \mathcal{F}$, closure under countable intersections, and closure under set difference.

Definition B.3 (Generated σ -algebra). Let $\mathcal{G} \subseteq \mathcal{P}(\Omega)$. The σ -algebra generated by $\mathcal{G}$ is

$$\sigma(\mathcal{G}) := \bigcap_{\mathcal{F} \text{ is a } \sigma\text{-algebra on } \Omega, \mathcal{G} \subseteq \mathcal{F}} \mathcal{F}$$

It is the smallest σ -algebra on Ω containing $\mathcal{G}$.

Definition B.4 (Borel σ -algebra). The *Borel σ -algebra* on $\mathbb{R}$ is

$$\mathcal{B}(\mathbb{R}) := \sigma(\{(a, b) \mid a < b, a, b \in \mathbb{R}\}).$$

Its elements are called *Borel sets*. More generally, $\mathcal{B}(\mathbb{R}^n)$ denotes the Borel σ -algebra on $\mathbb{R}^n$.

B.2 Probability spaces

Definition B.5 (Probability measure). Let $(\Omega, \mathcal{F})$ be a measurable space. A function $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$ is called a *probability measure* if:

1. $\mathbb{P}(\Omega) = 1$;
2. (σ -additivity) for every sequence $(A_n)_{n \in \mathbb{N}}$ of pairwise disjoint events,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Definition B.6 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where $(\Omega, \mathcal{F})$ is a measurable space and $\mathbb{P}$ is a probability measure on it.

The idea is the following:

- Ω is the *sample space*: the set of all possible outcomes.
- $\mathcal{F}$ is the σ -algebra of *events*: the subsets of Ω to which probabilities can be assigned.
- $\mathbb{P}$ assigns to each event its probability.

Proposition B.7 (Elementary properties of $\mathbb{P}$). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Then:

1. $\mathbb{P}(\emptyset) = 0$;
2. (*Monotonicity*) $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$;

3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
4. (*Countable sub-additivity*) $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$;
5. (*Continuity from below*) if $A_1 \subseteq A_2 \subseteq \dots$ then $\mathbb{P}(\bigcup_n A_n) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$.

Definition B.8 (Pushforward of probability measures). Let $(S_1, \mathcal{F}_1)$ and $(S_2, \mathcal{F}_2)$ be measurable spaces. We denote by $\mathfrak{P}(S, \mathcal{F})$ the *space of probability measures* on $(S, \mathcal{F})$, that is, the set of all probability measures defined on the measurable space $(S, \mathcal{F})$.

Given a measurable map $f : S_1 \rightarrow S_2$, the *pushforward* of a probability measure $\mu \in \mathfrak{P}(S_1, \mathcal{F}_1)$ is the probability measure $f_*\mu \in \mathfrak{P}(S_2, \mathcal{F}_2)$ defined by

$$f_*\mu(U) := \mu(f^{-1}(U)), \quad U \in \mathcal{F}_2.$$

This assignment is well defined since $f^{-1}(U) \in \mathcal{F}_1$ for every $U \in \mathcal{F}_2$ by measurability of f , and $f_*\mu$ is a probability measure because $f_*\mu(S_2) = \mu(f^{-1}(S_2)) = \mu(S_1) = 1$. The induced map

$$f_* : \mathfrak{P}(S_1, \mathcal{F}_1) \rightarrow \mathfrak{P}(S_2, \mathcal{F}_2), \quad \mu \mapsto f_*\mu$$

is called the *pushforward map*.

Definition B.9 (Dirac measure). Let $(X, \mathcal{F})$ be a measurable space and let $x \in X$. The *Dirac measure* (or *point mass*) at x is the map

$$\delta_x : \mathcal{F} \rightarrow [0, 1], \quad \delta_x(A) := \chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}, \quad A \in \mathcal{F}.$$

B.3 Random variables

Definition B.10 (Measurable function). Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. A function $f : \Omega_1 \rightarrow \Omega_2$ is *measurable* (with respect to $\mathcal{F}_1$ and $\mathcal{F}_2$) if

$$f^{-1}(B) := \{\omega \in \Omega_1 \mid f(\omega) \in B\} \in \mathcal{F}_1 \quad \forall B \in \mathcal{F}_2.$$

Definition B.11 (Random variable). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *random variable* is a measurable function

$$X : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

That is, X is a random variable if and only if

$$X^{-1}(B) = \{\omega \in \Omega \mid X(\omega) \in B\} \in \mathcal{F} \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

Remark B.12. The measurability condition ensures that statements such as $\{\omega : X(\omega) \leq x\}$ are genuine events, so that probabilities can be assigned to them.

Definition B.13 (Law of a random variable). The *law* (or *distribution*) of X is the probability measure $\mathbb{P}_X$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined by

$$\mathbb{P}_X(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B), \quad B \in \mathcal{B}(\mathbb{R}).$$

It is the *pushforward measure* of $\mathbb{P}$ along X .

B.4 Expected value

Definition B.14 (Expected value). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X a random variable. The *expected value* of X is defined as the Lebesgue integral

$$\mathbb{E}[X] := \int_{\Omega} X(\omega) d\mathbb{P}(\omega),$$

provided $\mathbb{E}[|X|] < +\infty$, in which case $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. By the transport formula,

$$\mathbb{E}[X] = \int_{\mathbb{R}} x d\mathbb{P}_X(x).$$

In the two standard cases:

- *Discrete*: $\mathbb{E}[X] = \sum_k x_k p_X(x_k)$;
- *Continuous*: $\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) dx$.

Definition B.15 (Independence of events). Events $A_1, \dots, A_n \in \mathcal{F}$ are *independent* if for every subset $I \subseteq \{1, \dots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

Definition B.16 (Independence of random variables). Random variables $X_1, \dots, X_n$ are *independent* if the generated σ -algebras $\sigma(X_1), \dots, \sigma(X_n)$ are independent, i.e., for every $B_1, \dots, B_n \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \prod_{i=1}^n \mathbb{P}(X_i \in B_i).$$

B.5 Ionescu-Tulcea theorem

Definition B.17. A *transition kernel* is a map

$$P : \mathcal{S} \times \mathcal{A} \times \mathcal{F}_S \rightarrow [0, 1]$$

satisfying:

- for every $(s, a) \in \mathcal{S} \times \mathcal{A}$, the map $B \mapsto P(B \mid s, a)$ is a probability measure on $(\mathcal{S}, \mathcal{F}_S)$;
- for every $B \in \mathcal{F}_S$, the map $(s, a) \mapsto P(B \mid s, a)$ is $\mathcal{F}_S \otimes \mathcal{P}(\mathcal{A})$ -measurable.

Theorem B.18 (Ionescu-Tulcea). Let $(\mathcal{S}, \mathcal{F}_S)$ and $(\mathcal{A}, \mathcal{P}(\mathcal{A}))$ be measurable spaces, and let $\mu_0 \in \mathfrak{P}(\mathcal{S}, \mathcal{F}_S)$ be an initial distribution. Let P be a transition kernel from $(\mathcal{S} \times \mathcal{A}, \mathcal{F}_S \otimes \mathcal{P}(\mathcal{A}))$ to $(\mathcal{S}, \mathcal{F}_S)$, and let $(\pi_t)_{t \in \mathbb{N}}$ be a sequence of transition kernels from $(\mathcal{S}, \mathcal{F}_S)$ to $(\mathcal{A}, \mathcal{P}(\mathcal{A}))$. Define the trajectory space

$$\Omega := \mathcal{S} \times \mathcal{A} \times \mathcal{S} \times \mathcal{A} \times \dots = \prod_{t \in \mathbb{N}} (\mathcal{S} \times \mathcal{A}),$$

equipped with the product σ -algebra

$$\mathcal{F} := \bigotimes_{t \in \mathbb{N}} (\mathcal{F}_S \otimes \mathcal{P}(\mathcal{A})).$$

Then there exists a unique probability measure $\mathbb{P}^{\mu_0, \pi}$ on $(\Omega, \mathcal{F})$ such that, for every $t \in \mathbb{N}$ and every $B \in \mathcal{F}_S$, $C \in \mathcal{P}(\mathcal{A})$:

$$\begin{aligned} \mathbb{P}^{\mu_0, \pi}(S_0 \in B) &= \mu_0(B), \\ \mathbb{P}^{\mu_0, \pi}(A_t \in C \mid S_t) &= \pi_t(C \mid S_t) \quad \mathbb{P}^{\mu_0, \pi}\text{-a.s.}, \\ \mathbb{P}^{\mu_0, \pi}(S_{t+1} \in B \mid S_t, A_t) &= P(B \mid S_t, A_t) \quad \mathbb{P}^{\mu_0, \pi}\text{-a.s.}, \end{aligned}$$

where $S_t, A_t : \Omega \rightarrow \mathcal{S}, \mathcal{A}$ denote the canonical projection maps

$$S_t(s_0, a_0, s_1, a_1, \dots) := s_t, \quad A_t(s_0, a_0, s_1, a_1, \dots) := a_t.$$

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